

Environmental change leaves no universal signature of phenotypic divergence

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1 Abstract

Environmental change can drive phenotypic divergence within species, but average divergence alone reveals neither whether population differences exceed variation among individuals nor whether the expected response generalizes across systems. We conducted a preregistered reanalysis of PROCEED v6.2, comprising 7,186 contrasts from 254 references and 253 species. We used $\ln M$ to quantify phenotypic differences between paired population samples relative to the variation within them and fitted phylogenetic location-scale meta-regressions to estimate both average separation and its predictability across systems. Across the dataset, there was no clear tendency for separation between paired samples to exceed within-population variation, while total heterogeneity reached 96.6%, providing little support for a universal magnitude of divergence. Disturbance contexts differed more strongly in predictability than in average separation. Landscape change and introduced populations produced greater separation than hunting or harvesting and *in situ* natural variation. Landscape change also produced comparatively predictable separation, whereas responses to hunting or harvesting, natural variation, and introduced species varied widely among systems. Although these moderator associations do not identify independent causal effects, they reveal that the biological usefulness of an average response depends on its predictability. Similar average divergence can represent either a broadly recurring response or a mixture of negligible and extreme outcomes. The comparatively predictable separation associated with landscape change may therefore support broader expectations across systems, whereas the variable outcomes of harvesting and responses to introduced species caution against transferring an average response among populations. In these less predictable contexts, local disturbance intensity, ecological interactions, and population history may be especially important. Treating heterogeneity as a biological result therefore defines the limits of generalization, revealing when broad syntheses can guide expectations and when population-specific evidence is essential.

1.1 Keywords

contemporary evolution; human disturbance; intraspecific variation; phenotypic heterogeneity; location-scale meta-analysis; global change

2 Introduction

Human activities reshape ecological conditions worldwide, altering the selective, developmental, and demographic environments experienced by wild populations (Alberti et al., 2017; Hendry et al., 2017). Within species phenotypic divergence has consequently been documented across diverse taxa and traits (Sanderson et al., 2022), but its magnitude varies considerably. Disturbances differ in intensity, duration, and spatial extent, while populations differ in their evolutionary histories and capacities for plastic and evolutionary change (Clement et al., 2023; Diamond and Martin, 2016; Turner, 2010). Previous syntheses found only small average differences in phenotypic change between human-disturbed and natural systems, alongside substantial variation among disturbance contexts and biological systems (Hendry et al., 2008; Sanderson et al., 2022). The central question is therefore not only how much populations diverge, but whether disturbances produce a predictable phenotypic signature across populations.

Variation around the average may itself contain important biological information. Similar mean divergence can arise because most populations respond by comparable amounts or because responses range from negligible to extreme (Sanderson et al., 2022, 2023). These alternatives may reflect different underlying processes and provide very different levels of predictability for conservation and management (Palkovacs et al., 2012; Roches et al., 2017). Moreover, the biological importance of a difference in trait means depends on the variation retained within the populations being compared: the same difference represents stronger phenotypic separation when within-population variation is low (Lynch, 1990; Nakagawa et al., 2026a). Yet conventional meta-regressions typically estimate moderator effects on the mean while assuming that residual heterogeneity is constant across disturbance contexts, traits, and study designs. If heterogeneity differs among these groups, imposing a common variance can conceal differences in cross-system predictability and change both estimated mean effects and their uncertainty (Nakagawa et al., 2025a; Yang et al., 2025). Previous conclusions may therefore be incomplete because they describe average divergence without establishing how reliably that average generalizes across systems.

Here, we reanalyse the Phenotypic Rates of Change Evolutionary and Ecological Database (PROCEED) (Gotanda and Gorné, 2026) following a preregistered protocol (Ortega et al., 2026). We quantify divergence using $\ln M$, the log ratio of phenotypic separation between paired population samples to variation within those samples, and fit multilevel phylogenetic location-scale meta-regressions. For each moderator, we ask whether the expected magnitude of separation differs among contexts and how predictable that magnitude is across contrasts. Our primary analysis compares six forms of human disturbance with *in situ* natural

variation. Additional models examine comparison design, elapsed years and generations, trait type, and study type, distinguishing field-observed phenotypic differences from estimates based on common-garden breeding or quantitative-genetic approaches. By modelling average divergence and residual heterogeneity jointly, we reassess both the magnitude of the phenotypic signature associated with human disturbance and the extent to which it generalizes across wild populations.

3 Methods

3.1 Data source and analytical contrasts

We used PROCEED version 6.2, a global compilation of quantitative within-species phenotypic comparisons (Gotanda and Gorné, 2026). We retained contrasts reporting the mean, standard deviation, and sample size for both samples and, following Sanderson et al. (2022), restricted the dataset to comparisons with known elapsed intervals of ≤ 300 generations. PROCEED distinguishes two comparison designs. Allochronic contrasts compare the same population or study system at different time points and therefore estimate phenotypic change along a temporal trajectory. Synchronic contrasts compare different populations or conditions sampled after a known divergence interval and therefore estimate phenotypic separation between them (Gotanda and Gorné, 2026). For allochronic contrasts, elapsed years represent the interval between the earlier and later samples; for synchronic contrasts, they represent the interval from population divergence to sampling. Elapsed generations were calculated by dividing elapsed years by species generation time. PROCEED orders samples chronologically or, when identifiable, from the source, ancestral, or original condition to the derived or novel condition. This ordering does not affect $\ln M$, which quantifies the magnitude rather than the direction of separation.

Disturbance context was the primary moderator. It distinguished climate change, hunting or harvesting, introduced populations, landscape change, pollution, and native responses to introduced species from contrasts representing *in situ* natural variation. Secondary models examined comparison design, elapsed time in years or generations, and trait type, classified as behaviour, body size, growth, other life-history, other morphology, phenology, physiology, or response traits. We also examined study type, distinguishing measurements of individuals that developed under natural conditions from estimates based on common-garden breeding or quantitative-genetic approaches. This classification did not imply that field-observed phenotypic differences were caused exclusively by plasticity or lacked a genetic basis. The primary models included 7,186 contrasts

⁸⁴ from 254 source references and 253 species (Table S1; Fig. [1](#)).



Figure 1: **Taxonomic representation of the 7,186 contrasts included in the analyses.** Rectangle area is proportional to the number of contrasts contributed by each taxonomic group; area therefore represents the amount of evidence rather than species richness. Labels report the corresponding numbers of contrasts and canonical species. Animal silhouettes were provided by Anna Lenz

3.2 Effect-size estimation

We quantified phenotypic divergence using the log-magnitude effect size, $\ln M$, which expresses phenotypic separation between two population samples relative to the variation within them (Nakagawa et al., 2026a). For two samples with means $\bar{x}_1$ and $\bar{x}_2$, standard deviations s_1 and s_2 , and sample sizes n_1 and n_2 , the between-group and pooled within-group mean squares are:

$$\begin{aligned} \text{MS}_B &= \frac{n_1 n_2}{n_1 + n_2} (\bar{x}_1 - \bar{x}_2)^2, \\ \text{MS}_W &= \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}. \end{aligned} \tag{1}$$

The pooled within-group variance and estimated between-group variance component are:

$$\begin{aligned} s_W^2 &= \text{MS}_W, \\ s_B^2 &= \frac{\text{MS}_B - \text{MS}_W}{n_0}, \quad n_0 = \frac{2n_1 n_2}{n_1 + n_2}, \end{aligned} \tag{2}$$

where n_0 is the harmonic mean of the two sample sizes. The effect size is then defined as:

$$\ln M = \ln \left(\frac{s_B}{s_W} \right). \tag{3}$$

Thus, $\ln M$ is the log ratio of the between-group, or mean-separation, standard deviation to the pooled within-group standard deviation. Positive values indicate that separation between the samples exceeds the variation within them, whereas negative values indicate that within-sample variation exceeds between-sample separation. Importantly, $\ln M = 0$ means that $s_B = s_W$, not that the two sample means are equal. Because MS_B depends on the squared difference between means, $\ln M$ quantifies the magnitude rather than the direction of phenotypic separation.

The analytical estimator becomes unstable as MS_B approaches MS_W and is undefined when $\text{MS}_B \leq \text{MS}_W$. We therefore used SAFE (Nakagawa et al., 2025b), a single-fit parametric bootstrap, to estimate $\ln M$ (y_i) and its sampling variance (v_i) for each contrast i (Supplementary material S1).

3.3 Phylogenetic location–scale meta-analysis

We built a phylogenetic scaffold from the Open Tree of Life (Hinchliff et al., 2015) using `rotl` [v3.1.1] (Michonneau et al., 2016) and `prepR4pcm` [v1.0.0] (Nakagawa et al., 2026b). Taxonomic reconciliation was performed across the complete PROCEED dataset rather than separately for each model. Within the analytical dataset, 261 of 267 submitted taxon names were resolved. Synonyms and infraspecific names, including subspecies, were then collapsed to their accepted species-level canonical names, yielding 253 unique species for the phylogeny-adjusted models. The six unresolved names, representing 51 contrasts, were excluded. We retrieved the induced subtree containing the resolved taxa and used `ape` [v5.8-0] (Paradis et al., 2004) to resolve multifurcations randomly into bifurcations with zero-length branches. When branch lengths were unavailable, we assigned Grafen branch lengths with power $\rho = 1$. Finally, we converted the resulting tree into a species-level phylogenetic correlation matrix under a Brownian-motion model of trait evolution using the using the function `prepR4pcm::pr_phylo_cor`.

3.4 Phylogenetic location–scale meta-regression

We fitted Bayesian multilevel location–scale meta-regressions using `brms` [v2.23.0] with `cmdstanr` [v0.9.0.9000] (Bürkner, 2017; Gabry et al., 2025; Nakagawa et al., 2025a; Stan Development Team, 2025) in R [v4.6.1] (R Core Team, 2026). We estimated $\ln M$ and its sampling variance using the `orchaRd` [v2.2.0] package. We analysed moderators separately due to differing data coverage and overlapping classifications (Table S). Each model included the focal moderator in both the location and scale components. We modelled the mean magnitude of phenotypic divergence (y_i) as:

$$y_i = \mu_i + u_i^{(\text{ref})} + a_i + \epsilon_i,$$

where

$$\mu_i = \beta_0^{(l)} + \sum_{k=1}^p \beta_k^{(l)} x_{ki}.$$

Here, μ_i is the expected mean $\ln M$, $\beta_0^{(l)}$ is the location intercept, and $\beta_k^{(l)}$ is the coefficient for moderator x_k . The terms $u_i^{(\text{ref})}$ and a_i represent random effects for source reference and phylogenetic species, respectively. We modelled the residual term as:

$$\epsilon_i \sim N(0, v_i + \sigma_i^2),$$

where v_i is the known sampling variance and σ_i^2 is the residual heterogeneity. To test if residual heterogeneity differed among contexts, we modelled the residual standard deviation as:

$$\log \sigma_i = \beta_0^{(s)} + \sum_{m=1}^q \beta_m^{(s)} z_{mi}.$$

Positive $\beta_m^{(s)}$ values indicate greater residual heterogeneity (i.e., less predictable magnitude of divergence between groups). We estimated location and scale components jointly. A sensitivity analysis included study system as an additional random effect (Table S).

3.5 Posterior inference, heterogeneity, robustness, and reproducibility

We based inference for continuous moderators on posterior slopes and for categorical moderators on population-level marginal means and pairwise contrasts. We report posterior means with 95% highest-posterior-density credible intervals (CrI) and treated intervals excluding zero as well supported. Using the intercept-only model, we calculated total and component-specific I^2 values to partition heterogeneity among source references, phylogenetically structured species effects, and residual variation (Nakagawa et al., 2025a). Models were fitted using four chains of 4,000 iterations, including 2,000 warm-up iterations, and convergence was assessed from $\hat{R}$, effective sample sizes, divergent transitions, and maximum tree-depth warnings (Vehtari et al., 2021). Robustness was evaluated using three prespecified sensitivity analyses: restricting the data to contrasts with a combined sample size of at least 40, removing the phylogenetic random effect, and adding a study-system random effect (Supplementary Methods). We additionally assessed small-study effects using the half harmonic-mean sample size, $\tilde{n}_0 = n_0/2$, where n_0 was defined above. Following the modified Egger-type approach developed for $\ln M$ (Nakagawa et al., 2026a), we used $1/\sqrt{\tilde{n}_0}$ as a sample-size-based precision predictor rather than the sampling standard error, because the latter is structurally correlated with $\ln M$. We included $1/\sqrt{\tilde{n}_0}$ in both the location and scale components: the location slope tested whether mean $\ln M$ varied systematically with study size, whereas the scale slope tested whether residual heterogeneity, and hence the predictability of phenotypic separation, varied with study size. These analyses were treated as diagnostics of small-study effects rather than direct tests of publication bias (Egger et al., 1997; Nakagawa et al., 2022).

We followed the PRISMA-EcoEvo (O’Dea et al., 2021) reporting guidance and provide the completed checklist as supplementary material. All analytical code is available at https://github.com/Santiago-Ortega/Magnitude_pace_modern_life, and a step-by-step reproducible account of data preparation, effect-size es-

timation, model fitting, diagnostics, sensitivity analyses, and figure generation is available at https://santiago-Ortega.github.io/Magnitude_pace_modern_life/. Figures were produced in R using `ggplot2` [v4.03] (Wickham, 2011), `ggbeeswarm` [v0.7.3] (Clarke et al., 2025), and `patchwork` [v1.3.2] (Pender-
sen, 2025). To aid interpretation, we converted the location estimates to approximate d -equivalent values, d_{eq} (Nakagawa et al., 2026a). These values are reported in the text and displayed on secondary axes in the location panels; pairwise contrasts and regression slopes were retained on the $\ln M$ scale because they represent differences or changes rather than separation. On the conventional standardized mean-difference scale, values of $d = 0.2$, 0.5 , and 0.8 are commonly described as small, medium, and large, respectively (Cohen, 1988; Lakens, 2013). By comparison, $\ln M = 0$ corresponds approximately to $d_{eq} = 1.41$, indicating substantial separation between group means relative to within-group variation.

4 Results

4.1 Data overview

Most contrasts compared populations or samples measured synchronically (64.5%, $n = 4,635$), whereas the remaining 35.5% were allochronic comparisons ($n = 2,551$). Introductions were by far the most frequently represented disturbance context (57.1%, $n = 4,106$). Other *in situ* natural variation (18.7%, $n = 1,345$) and hunting or harvesting (12.7%, $n = 913$) were the next most common contexts. Climate change, landscape change, responses to introduced species, and pollution each contributed less than 4% of the available contrasts. Comparison design was strongly associated with disturbance context (Cramér’s $V = 0.896$; Fig. S, indicating that synchronic and allochronic contrasts were not distributed independently across contexts. Studies of phenotypic differences expressed in wild populations predominated (63.9%, $n = 4,592$), while common-garden, quantitative-genetic, and related designs intended to distinguish heritable divergence from environmentally induced variation accounted for 36.1% of the dataset ($n = 2,594$; Fig. 2). The evidence base was also strongly concentrated in morphological traits: body size (37.4%, $n = 2,684$) and other morphological traits (33.5%, $n = 2,410$) together accounted for 70.9% of the analytical sample. Other life-history traits represented a further 18.2% ($n = 1,308$), whereas physiological, growth, phenological, performance, and behavioural traits were comparatively uncommon.

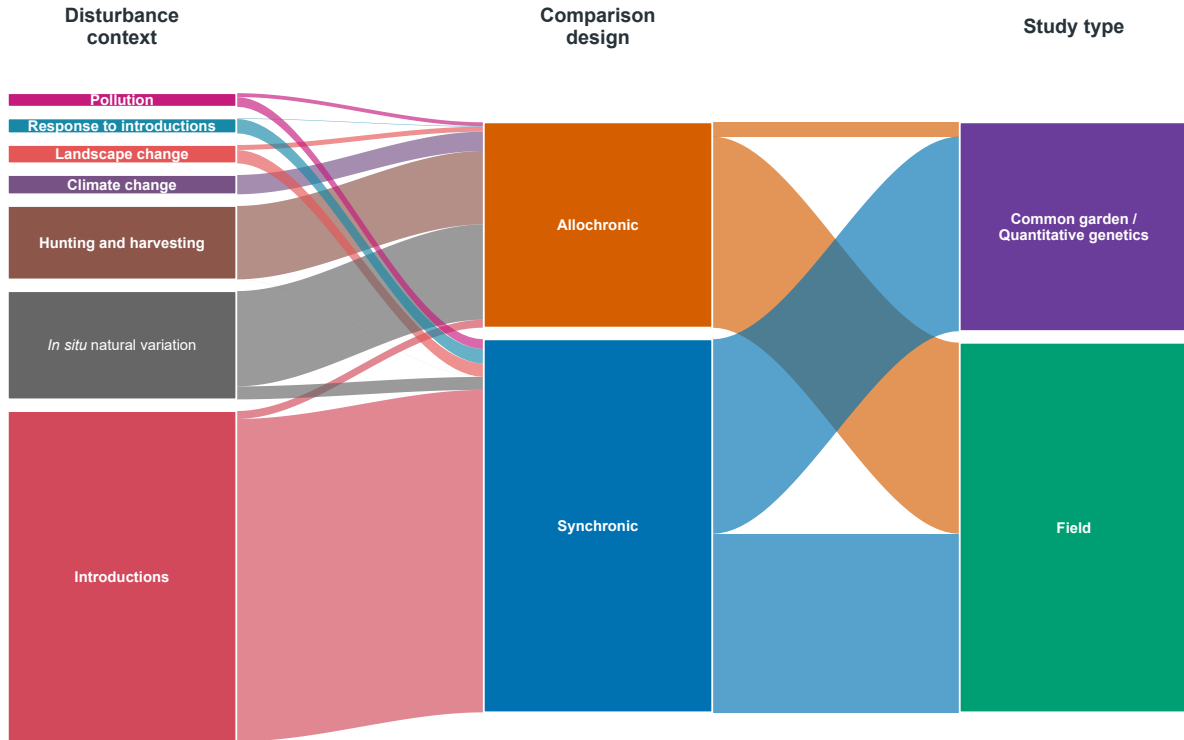


Figure 2: **Structure of the evidence base across comparison designs, study types, and disturbance contexts.** The alluvial diagram represents the 7,186 contrasts included in analyses. The height of each block and the width of each connecting ribbon are proportional to the number of contrasts. Ribbons show how synchronic and allochronic comparisons were distributed between field-based studies and common-garden or quantitative-genetic studies, and subsequently across disturbance contexts: introductions involving comparisons of introduced populations, *in situ* natural variation, hunting or harvesting, climate change, landscape change, responses of other species to introductions, and pollution.

4.2 Overall magnitude and predictability of phenotypic divergence

Across the dataset, there was no clear tendency for phenotypic separation between paired samples to exceed within-population variation ($\beta_{\text{Overall}}^{(l)} = -0.55$, 95% CrI: $[-1.41, 0.30]$). Nevertheless, the overall estimate corresponded to an approximate standardized mean difference of $d_{\text{eq}} = 0.81$ (95% CrI: $[0.34, 1.92]$), indicating large separation between the paired samples on a standardized mean-difference scale. The magnitude of separation varied substantially among contrasts ($\beta_{\text{Overall}}^{(s)} = 0. - 59$, 95% CrI: $[-0.61, -0.56]$), resulting in low overall predictability across comparisons (Fig. 3). Furthermore, we detected a small-study effect in the location component ($\beta_{1/\sqrt{n_0}}^{(l)} = 1.75$, 95% CrI: $[1.55, 1.95]$), indicating that smaller studies tended to report larger $\ln M$ values. Small-study effects were also evident in the scale component ($\beta_{1/\sqrt{n_0}}^{(s)} = -1.37$, 95% CrI: $[-1.60, -1.15]$), with smaller studies showing lower residual heterogeneity. Thus, the absolute $\ln M$ values reported below should be interpreted cautiously, as separation may be somewhat weaker relative to within-population variation in larger studies. Total heterogeneity was $I_{\text{total}}^2 = 96.6\%$ (95% CrI: $[95.5, 97.7]$), comprising variation attributable to source references (37.3%, $[23.8, 52.4]$), phylogenetically structured species effects (38.1%, $[16.2, 58.9]$), and residual differences among contrasts (21.0%, $[14.6, 28.4]$).

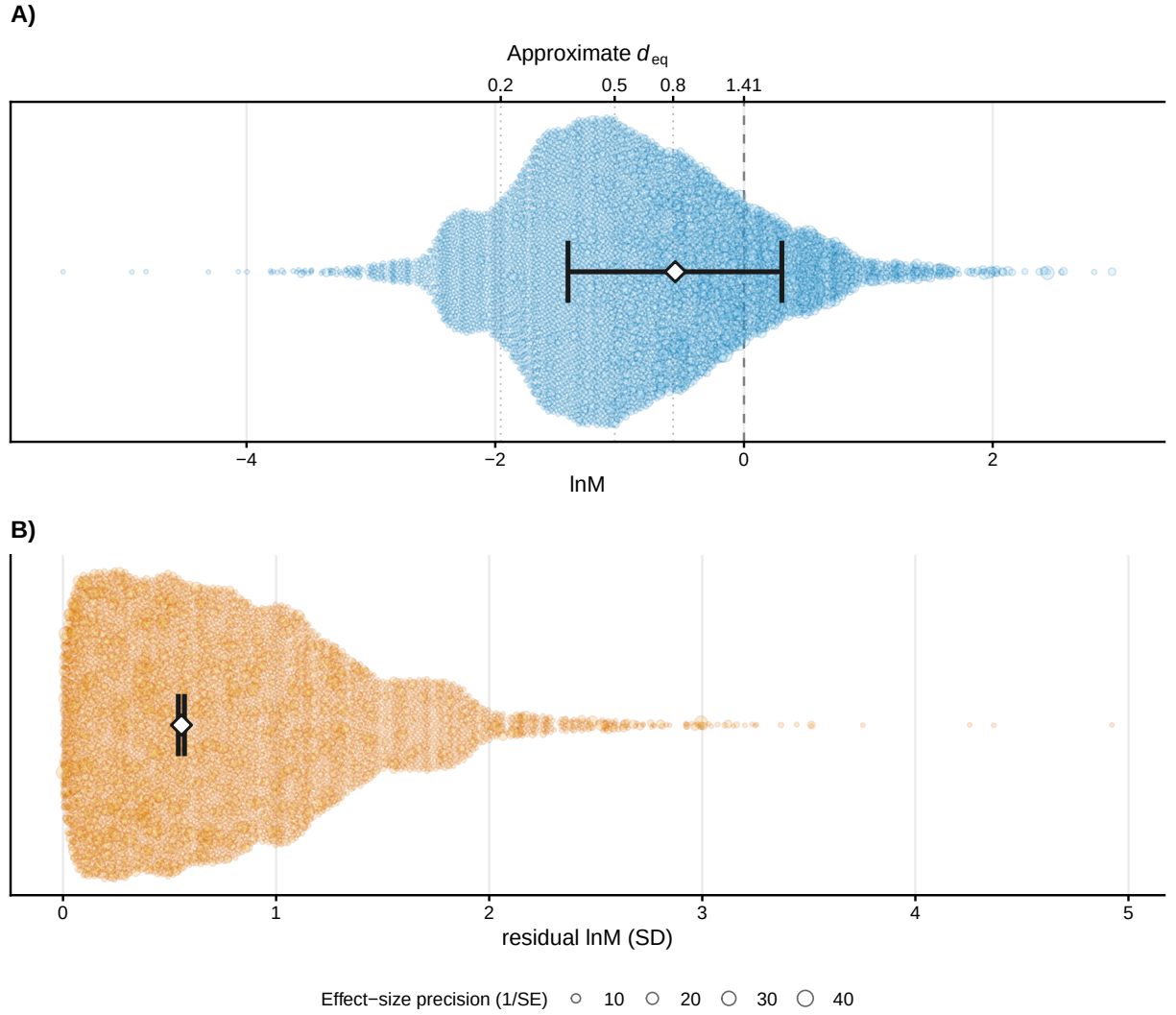


Figure 3: Overall magnitude and predictability of within-species phenotypic divergence across 7,186 contrasts. **(A)** Average magnitude of separation between paired samples. The dashed line marks $\ln M = 0$ (approximately $d_{eq} = 1.41$), indicating substantial separation between group means; the upper axis provides approximate d_{eq} values. **(B)** Predictability in the magnitude of separation among contrasts, quantified by the residual standard deviation, σ . Larger σ values indicate lower predictability. White circles and black lines denote posterior estimates and 95% credible intervals, respectively, while bubble size indicates precision ($1/SE$).

4.3 Disturbance context

Among disturbance contexts, hunting or harvesting and *in situ* natural variation were the only contexts in which phenotypic separation between samples was clearly smaller than variation within populations (hunting or harvesting: $\beta_{\text{Hunting}}^{(l)} = -0.80$, 95% CrI: $[-1.61, -0.04]$; $\beta_{\text{Natural}}^{(l)} = -0.81$, $[-1.61, -0.06]$; Fig. 4A). Nevertheless, both corresponded to moderate standardized separation between sample means (hunting or harvesting: $d_{\text{eq}} = 0.64$, $[0.28, 1.36]$; natural variation: $d_{\text{eq}} = 0.63$, $[0.28, 1.33]$). For the remaining contexts, evidence was inconclusive as to whether between-sample separation was greater or smaller than within-sample variation. Pairwise comparisons nevertheless showed that landscape change produced greater separation than hunting or harvesting ($\beta_{\text{Landscape-Hunting}}^{(l)} = 0.483$, 95% CrI: $[0.132, 0.827]$) and *in situ* natural variation ($\beta_{\text{Landscape-Natural}}^{(l)} = 0.488$, $[0.160, 0.836]$). Introductions likewise showed greater separation than hunting or harvesting ($\beta_{\text{Introduction-Hunting}}^{(l)} = 0.282$, $[0.035, 0.550]$) and *in situ* natural variation ($\beta_{\text{Introduction-Natural}}^{(l)} = 0.285$, $[0.043, 0.541]$). Evidence for the remaining pairwise location contrasts was inconclusive.

Predictability in the magnitude of separation differed more broadly among disturbance contexts (Fig. 4 B). Hunting or harvesting showed greater residual variation, and therefore lower predictability, than climate change ($\beta_{\text{Hunting-Climate}}^{(s)} = 0.365$, 95% CrI: $[0.186, 0.550]$), introductions ($\beta_{\text{Hunting-Introduction}}^{(s)} = 0.350$, $[0.278, 0.420]$), natural variation ($\beta_{\text{Hunting-Natural}}^{(s)} = 0.227$, $[0.144, 0.311]$), landscape change ($\beta_{\text{Hunting-Landscape}}^{(s)} = 0.796$, $[0.558, 1.047]$), and pollution ($\beta_{\text{Hunting-Pollution}}^{(s)} = 0.549$, $[0.294, 0.804]$). Responses to introductions showed lower predictability than climate change ($\beta_{\text{Response-Climate}}^{(s)} = 0.256$, $[0.032, 0.489]$), introductions ($\beta_{\text{Response-Introduction}}^{(s)} = 0.241$, $[0.085, 0.386]$), landscape change ($\beta_{\text{Response-Landscape}}^{(s)} = 0.687$, $[0.406, 0.962]$), and pollution ($\beta_{\text{Response-Pollution}}^{(s)} = 0.440$, $[0.163, 0.742]$). Natural variation showed lower predictability than introductions ($\beta_{\text{Natural-Introduction}}^{(s)} = 0.123$, $[0.053, 0.190]$), landscape change ($\beta_{\text{Natural-Landscape}}^{(s)} = 0.570$, $[0.334, 0.819]$), and pollution ($\beta_{\text{Natural-Pollution}}^{(s)} = 0.322$, $[0.066, 0.581]$). Climate change ($\beta_{\text{Climate-Landscape}}^{(s)} = 0.431$, $[0.148, 0.742]$) and introductions ($\beta_{\text{Introduction-Landscape}}^{(s)} = 0.447$, $[0.222, 0.703]$) were also less predictable than landscape change.

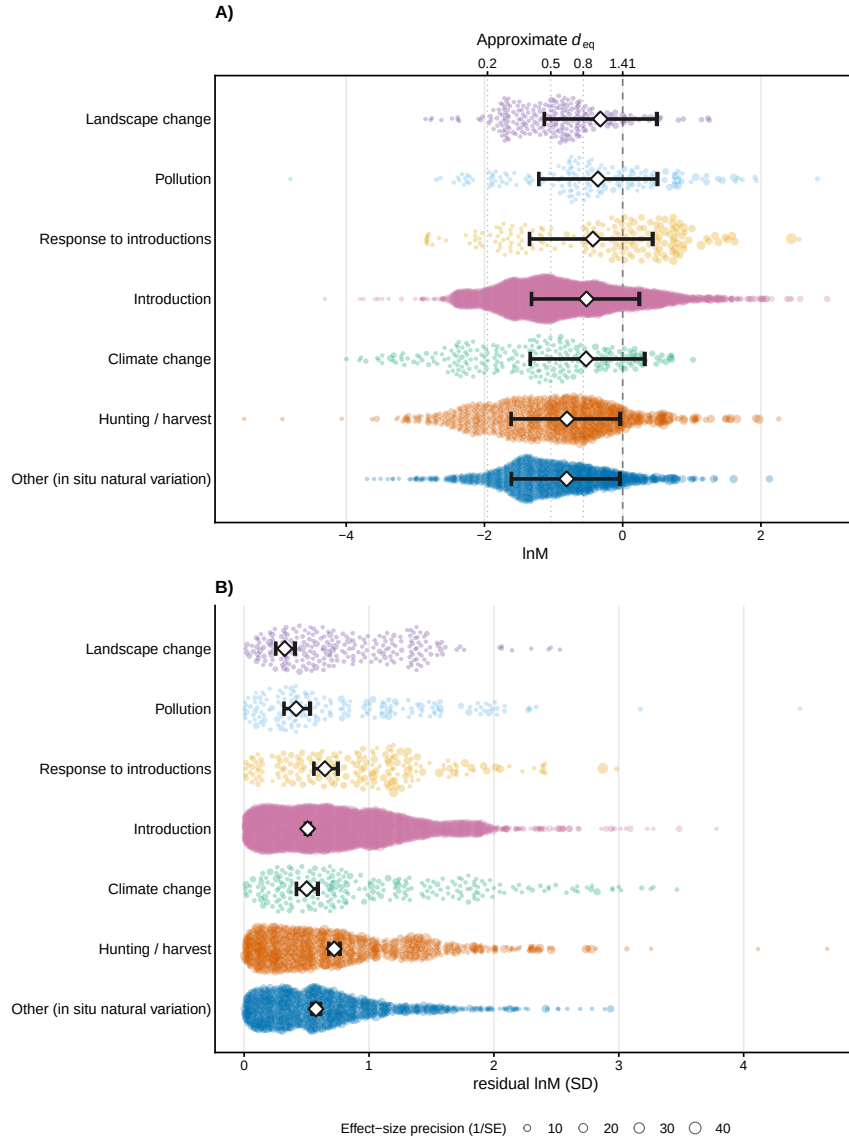


Figure 4: **Magnitude and predictability of within-species phenotypic divergence across disturbance contexts.** (A) Mean phenotypic separation between paired samples for each context. The reference line at $\ln M = 0$ corresponds to an absolute standardized mean difference of approximately $d_{eq} = 1.41$; the upper axis translates $\ln M$ into these approximate values. (B) Predictability of this separation across contrasts within each context, where a larger σ indicates lower predictability. White circles and black lines represent posterior estimates and 95% credible intervals, respectively. Bubble size is proportional to precision (1/SE).

4.4 Other moderator analyses

Comparison design was associated with both the magnitude and predictability of phenotypic divergence. For synchronic contrasts, average separation between populations was clearly smaller than the phenotypic variation retained within them. Allochronic contrasts showed a similar pattern, although the evidence relative to $\ln M = 0$ was marginal. Despite this shared tendency, synchronic contrasts showed greater average separation than allochronic contrasts ($\beta_{\text{Synchronic-Allochronic}}^{(l)} = 0.251$, 95% CrI: [0.081, 0.420]). They also showed lower residual variation in separation magnitude ($\beta_{\text{Synchronic-Allochronic}}^{(s)} = -0.208$, 95% CrI: [-0.260, -0.157]), indicating greater predictability.

Elapsed time was positively associated with average separation and negatively associated with residual variation among contrasts (Fig. 5). A tenfold increase in elapsed years was associated with a 36% increase in the ratio of between-group separation to within-group variation ($\beta_{\log_{10}(\text{years})}^{(l)} = 0.307$, 95% CrI: [0.235, 0.381]) and a 15% reduction in the residual standard deviation ($\beta_{\log_{10}(\text{years})}^{(s)} = -0.167$, [-0.215, -0.119]). Elapsed generations produced a similar pattern: a tenfold increase was associated with a 33% increase in the separation ratio ($\beta_{\log_{10}(\text{generations})}^{(l)} = 0.286$, [0.215, 0.359]) and a 14% reduction in the residual standard deviation ($\beta_{\log_{10}(\text{generations})}^{(s)} = -0.149$, [-0.182, -0.117]).

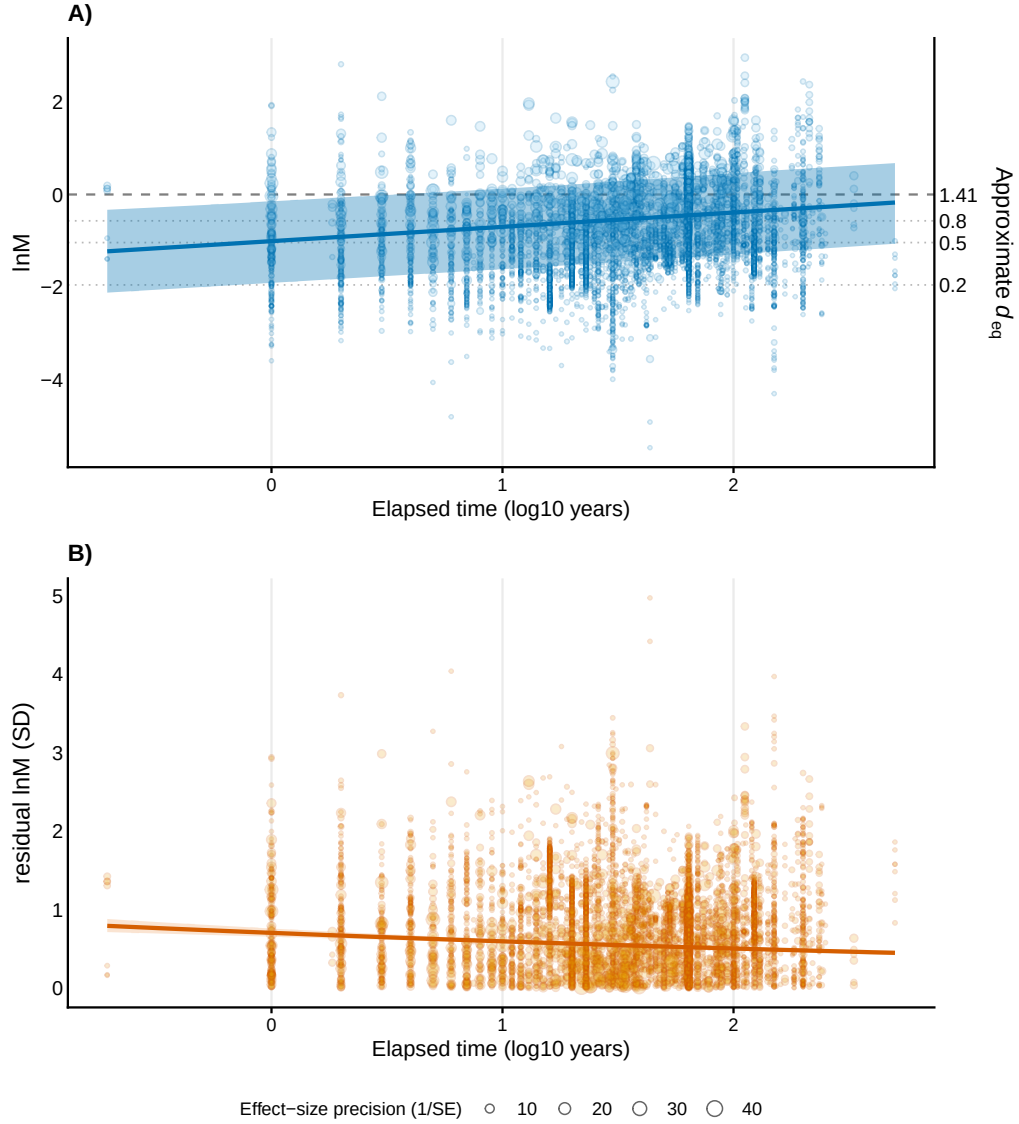


Figure 5: **Elapsed time is associated with the magnitude and predictability of within-species phenotypic divergence.** (A) Average separation between paired samples over elapsed time (log₁₀ years). The reference line at $\ln M = 0$ corresponds to an absolute standardized mean difference of approximately $|d| = 1.41$, which is also indicated on the right axis. (B) Predictability of separation among contrasts over elapsed time. A larger σ indicates lower predictability. The solid line and shaded ribbon show the posterior estimate and 95% credible interval, respectively. One horizontal unit represents a tenfold increase in time, and bubble size reflects precision (1/SE).

218 Trait types also differed in their average magnitude and predictability of divergence (Fig. 6 A; Table S).
219 Behavioural traits were the only category for which mean $\ln M$ was clearly below zero, indicating that
220 average separation between samples was smaller than the phenotypic variation retained within them. For
221 the remaining trait types, evidence was inconclusive as to whether between-sample separation exceeded
222 within-sample variation. Pairwise comparisons showed that behavioural traits had lower average separation
223 than growth, other life-history, response, and body-size traits. Other life-history and response traits also
224 showed greater separation than several morphological and physiological categories. Differences were more
225 extensive in the scale component (Fig. 6 B). Behavioural and body-size traits were among the least predictable
226 categories, whereas growth traits produced comparatively predictable estimates. Other life-history traits
227 also showed greater residual variation than other morphological, growth, and physiological traits. Complete
228 pairwise contrasts for the location and scale components are reported in Table S.

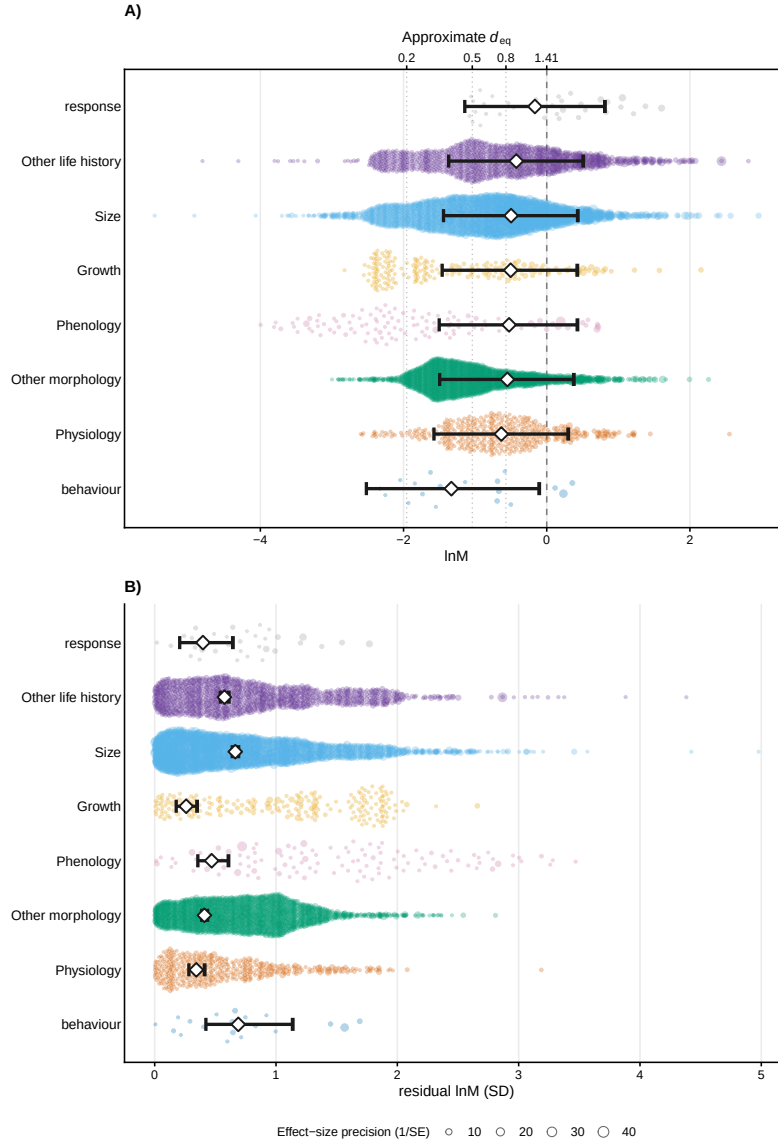


Figure 6: **Magnitude and predictability of within-species phenotypic divergence across trait types.** (A) Average phenotypic separation between paired population samples relative to within-population variation for each trait type. The upper axis displays approximate absolute standardized mean differences. (B) Predictability of separation among contrasts within each trait type. White circles and black intervals represent posterior estimates and 95% credible intervals, respectively. Bubble size indicates precision (1/SE). Response traits denote measures of performance expressed as ratios.

Study type was similarly associated with both model components. For field-based contrasts involving individuals that developed under natural conditions, mean $\ln M$ indicated that average separation between samples was smaller than the phenotypic variation retained within them. For contrasts based on common-garden breeding, quantitative-genetic methods, or related approaches, evidence was inconclusive as to whether separation exceeded within-group variation. Nevertheless, these controlled-study contrasts showed greater average separation than field-based contrasts ($\beta_{\text{Common garden/QG-Field}}^{(l)} = 0.211$, 95% CrI: [0.062, 0.353]). They also showed lower residual variation ($\beta_{\text{Common garden/QG-Field}}^{(s)} = -0.338$, 95% CrI: [-0.394, -0.275]), indicating greater predictability.

4.5 Excluded analyses and sensitivity analyses

Four preregistered moderator analyses were excluded because they did not meet the convergence criteria of $\hat{R} \leq 1.01$, effective sample sizes ≥ 400 , and no divergent transitions (Vehtari et al., 2021). These models tested whether the average magnitude and predictability of separation differed among broad taxonomic groups, mathematical forms of trait measurement, transformed and untransformed trait values, and ongoing versus novel environmental change; their estimates were not interpreted.

Most conclusions were stable across the 21 prespecified sensitivity models, particularly those for comparison design, elapsed time, study type, and disturbance-context location effects. Changes were concentrated in trait-type location contrasts and a small number of predictability effects. Departures under the minimum-sample-size and study-system specifications were interpreted cautiously because these models produced divergent transitions. Model-specific exclusions, coefficient-level comparisons, and convergence diagnostics are reported in the online supplement (https://santiago-Ortega.github.io/Magnitude_pace_modern_life/).

5 Discussion

5.1 Phenotypic divergence is heterogeneous rather than universal

We asked whether environmental change leaves a general and predictable signature of within-species phenotypic divergence. Across 7,186 contrasts, average phenotypic separation between paired population samples, measured by $\ln M$, did not clearly exceed within-population phenotypic variation (Fig. 3A). This does not mean that population means were identical. Rather, phenotypic differences between population samples did

not consistently outweigh variation among individuals within populations. This result complements previous syntheses showing that contemporary phenotypic change is widespread but highly variable across taxa, traits, and environmental contexts (Hendry et al., 2008; Sanderson et al., 2022, 2023).

The clearest pattern was the extent of this variation (Fig. 3B). Total heterogeneity exceeded 96%, and residual variation in $\ln M$ differed among disturbance contexts, comparison designs, elapsed intervals, trait types, and study types. Similar average separation can therefore reflect either comparable differences across systems or a mixture of weak and extreme responses (Sanderson et al., 2023). By modelling both the expected $\ln M$ and its residual variation, the location-scale models showed not only where phenotypic separation tended to be greater relative to within-population variation, but also where it was more predictable across the evidence base (Nakagawa et al., 2025a).

5.2 Disturbance context shapes the magnitude and predictability of divergence

Disturbance contexts differed in the average degree of phenotypic separation between population samples relative to within-population variation. Landscape-change contrasts showed greater separation than hunting or harvesting and *in situ* natural variation, while introduced populations also showed greater separation than these two contexts (Fig. 4A). This complements previous syntheses based on rates of phenotypic change (Hendry et al., 2008; Sanderson et al., 2022). A population mean can change rapidly without producing strong phenotypic separation when variation among individuals remains large. Rates of change and $\ln M$ therefore describe related but distinct features of population responses: how rapidly traits change and how distinct population samples become relative to the variation retained within populations.

Differences among disturbance contexts were even clearer in the predictability of separation (Fig. 4B). Landscape change produced comparatively predictable separation, whereas hunting or harvesting, *in situ* natural variation, and responses of resident populations to introduced species showed much greater variation among contrasts. Introduced populations themselves were intermediate, emphasizing that divergence following an introduction differs from the response of resident populations exposed to an introduced species. Thus, disturbance categories with comparable average separation can differ substantially in how well that average generalizes across populations.

Several biological mechanisms could generate these differences. Harvesting may impose strong selection, but targeted traits, selection direction and intensity, demographic effects, and management histories vary

among systems (Campbell-Staton et al., 2021; Conover and Munch, 2002; Festa-Bianchet, 2017). Responses to introduced species may likewise depend on whether the introduced organism acts as a predator, competitor, parasite, host, or prey, and on its ecological overlap with the focal population (Strauss et al., 2006). Landscape change, in contrast, can create persistent differences in habitat structure, connectivity, cover, and resource distribution (Fischer and Lindenmayer, 2007; Smokorowski and Pratt, 2007), potentially generating more repeatable degrees of phenotypic separation across populations. These mechanisms remain hypotheses, however, because our models describe differences among disturbance contexts rather than the processes generating them.

5.3 Temporal, biological, and methodological correlates of divergence

Comparison design and elapsed time captured related but different aspects of phenotypic divergence. Synchronic contrasts showed greater and more predictable separation than allochronic contrasts. Differences between populations may integrate their distinct environmental and demographic histories since divergence, whereas comparisons of the same population through time may be more sensitive to short-term changes in climate, density, age structure, demography, and individual responses (Crozier et al., 2011; Cunningham et al., 2020; Garant et al., 2004; Ozgul et al., 2010). However, comparison design was strongly associated with disturbance context (Cramér’s $V = 0.896$; Fig.S). The design effect therefore describes a pattern in the available evidence and should not be interpreted as an independent test of spatial versus temporal divergence.

Across both designs, longer elapsed intervals were associated with greater and more predictable separation (Fig. 5). Similar results were obtained when time was expressed in years or generations. This pattern is consistent with persistent environmental differences becoming clearer as plastic responses, demographic turnover, and selection act over longer periods (Crozier et al., 2011; Losos et al., 1997; Ozgul et al., 2010; Reznick et al., 1997). However, years and generations are closely related measures, so their agreement is supportive rather than an independent replication. Moreover, these models do not estimate a causal rate of divergence. Elapsed time describes the interval between endpoints or since population divergence, not the path of change between them, and it was unevenly distributed across disturbance contexts and study designs.

Trait type and study type further showed how biological variation and environmental control influence $\ln M$. Behavioural traits had low average separation and some of the least predictable estimates (Fig. 6). Although behaviour can be repeatable, it can also change rapidly with environmental conditions, social

context, and individual state, and individuals may differ in their responses to these conditions (Bell et al., 2009; Dingemanse and Wolf, 2013; Dingemanse et al., 2010). Behavioural contrasts may therefore combine persistent population differences with short-term variation at the time of measurement. Growth traits, by contrast, produced the most predictable estimates. Growth integrates physiological and environmental conditions over a developmental period and is limited by trade-offs among energy acquisition, maintenance, development, and reproduction (Dmitriew, 2011). These shared constraints may restrict the range of viable responses, although this explanation was not tested directly.

A related pattern appeared across study types. Field-based contrasts showed smaller and less predictable separation than common-garden and quantitative-genetic contrasts. Individuals developing in natural conditions experience variation in climate, food, density, age, condition, and developmental history, which can maintain substantial variation within populations (Gienapp et al., 2008; Merilä and Hendry, 2014). Common-garden and quantitative-genetic approaches standardize or model some of this variation, making persistent differences easier to detect. Because $\ln M$ measures between-group separation relative to within-group variation, reducing environmentally induced variation within groups can itself produce larger and more predictable estimates. The result therefore does not show that genetic divergence is generally stronger than divergence observed in the field; it may partly reflect differences in environmental control and in the systems represented by each study type.

5.4 Limitations and future opportunities

Several limitations provide context for these findings. First, disturbance context, comparison design, trait type, study type, and elapsed time were closely related within the dataset. Their effects could not be fully separated, so each model describes a pattern in the available evidence rather than an independent causal effect. Larger and more balanced datasets could support models that separate these influences. Second, the evidence was unevenly distributed across contexts, designs, traits, and taxa, with introductions, synchronic comparisons, morphological traits, and a few taxonomic groups particularly well represented (Figs. 1 and 2). More studies of under-represented systems are needed to determine how broadly the patterns apply. Third, most records compared only two endpoints. Long-term studies that repeatedly measure both trait means and variation among individuals are needed to distinguish gradual accumulation from short-term plasticity, abrupt shifts, and reversals.

Small-study effects provide an additional reason for caution. Smaller studies tended to report larger $\ln M$

values but lower residual heterogeneity, suggesting that they may portray phenotypic separation as both stronger and more predictable than larger studies (Nakagawa et al., 2022). These patterns do not demonstrate publication bias, because small-study effects can also arise from differences in study design, taxa, traits, or biological systems, but they suggest that the absolute magnitude and predictability of separation reported here may be somewhat overstated.

Finally, several preregistered moderator models did not meet our convergence criteria, and the phylogenetic reconstruction required approximations for unresolved relationships and branch lengths. Nevertheless, conclusions were generally stable across the 21 prespecified sensitivity models, particularly for comparison design, elapsed time, study type, and the average differences among disturbance contexts. Changes were concentrated in trait-type comparisons and a small number of predictability estimates. Some departures under the minimum-sample-size and study-system specifications also occurred in models with divergent transitions and should therefore be interpreted cautiously. Our findings support broad comparative patterns but should not be treated as precise predictions for particular species or systems.

5.5 Conclusions

Phenotypic divergence under environmental change is neither uniformly large nor uniformly predictable. Similar average differences can arise from broadly repeatable responses or from mixtures of weak and extreme divergence, making an average response a limited guide to what should be expected in any particular population. By combining $\ln M$ with phylogenetic location-scale meta-regression, we showed that disturbance context, elapsed time, comparison design, trait type, and study type were associated with both the magnitude and predictability of phenotypic separation. Studies of global change should therefore retain and report within-population variation, while syntheses should treat heterogeneity as information about how reliably responses generalize across systems rather than simply as statistical noise. Understanding and anticipating phenotypic responses requires knowing not only how much populations are expected to differ, but also how confidently that expectation can be extended to another population.

References

Marina Alberti, Cristian Correa, John M. Marzluff, Andrew P. Hendry, Eric P. Palkovacs, Kiyoko M. Gotanda, Victoria M. Hunt, Travis M. Apgar, and Yuyu Zhou. Global urban signatures of pheno-

typic change in animal and plant populations. *Proceedings of the National Academy of Sciences of the United States of America*, 114:8951–8956, 8 2017. ISSN 10916490. doi: 10.1073/PNAS.1606034114.

Alison M. Bell, Shala J. Hankison, and Kate L. Laskowski. The repeatability of behaviour: a meta-analysis. *Animal Behaviour*, 77:771–783, 4 2009. ISSN 0003-3472. doi: 10.1016/J.ANBEHAV.2008.12.022.

Paul Christian Bürkner. brms: An r package for bayesian multilevel models using stan. *Journal of Statistical Software*, 80:1–28, 8 2017. ISSN 1548-7660. doi: 10.18637/JSS.V080.I01. URL <https://www.jstatsoft.org/index.php/jss/article/view/v080i01>.

Shane C. Campbell-Staton, Brian J. Arnold, Dominique Gonçalves, Petter Granli, Joyce Poole, Ryan A. Long, and Robert M. Pringle. Ivory poaching and the rapid evolution of tusklessness in african elephants. *Science*, 374:483–487, 10 2021. ISSN 10959203. doi: 10.1126/SCIENCE.ABE7389.

Erik Clarke, Scott Sherrill-Mix, and Charlotte Dawson. *ggbeeswarm: Categorical Scatter (Violin Point) Plots*, 2025. URL <https://github.com/eclarke/ggbeeswarm>. R package version 0.7.3.

Dale T. Clement, Isabelle P. Neylan, Nicholas J. Roberts, Sebastian J. Schreiber, Pete C. Trimmer, and Andrew Sih. Evolutionary history mediates population response to rapid environmental change through within-generational and transgenerational plasticity. *The American Naturalist*, 201(5):E90–E109, 2023. doi: 10.1086/723624.

Jacob Cohen. *Statistical power analysis for the behavioral sciences*, Rev. Lawrence Erlbaum Associates, Inc, 1988.

David O. Conover and Stephan B. Munch. Sustaining fisheries yields over evolutionary time scales. *Science*, 297:94–96, 7 2002. ISSN 00368075. doi: 10.1126/SCIENCE.1074085.

Lisa G. Crozier, Mark D. Scheuerell, and Richard W. Zabel. Using time series analysis to characterize evolutionary and plastic responses to environmental change: A case study of a shift toward earlier migration date in sockeye salmon. *American Naturalist*, 178:755–773, 12 2011. ISSN 00030147. doi: 10.1086/662669.

George D. Cunningham, Geoffrey M. While, Mats Olsson, Gabriella Ljungström, and Erik Wapstra. Degrees of change: between and within population variation in thermal reaction norms of phenology in a viviparous lizard. *Ecology*, 101, 10 2020. ISSN 1939-9170. doi: 10.1002/ECY.3136.

Sarah E. Diamond and Ryan A. Martin. The interplay between plasticity and evolution in response to human-induced environmental change. *F1000Research*, 5:2835, 2016. doi: 10.12688/f1000research.9731.1.

- Niels J. Dingemanse and Max Wolf. Between-individual differences in behavioural plasticity within populations: causes and consequences. *Animal Behaviour*, 85:1031–1039, 5 2013. ISSN 0003-3472. doi: 10.1016/J.ANBEHAV.2012.12.032.
- Niels J. Dingemanse, Anahita J.N. Kazem, Denis Réale, and Jonathan Wright. Behavioural reaction norms: animal personality meets individual plasticity. *Trends in Ecology & Evolution*, 25:81–89, 2 2010. ISSN 0169-5347. doi: 10.1016/J.TREE.2009.07.013.
- Caitlin M. Dmitriew. The evolution of growth trajectories: What limits growth rate? *Biological Reviews*, 86:97–116, 2 2011. ISSN 14647931. doi: 10.1111/J.1469-185X.2010.00136.X. URL <https://onlinelibrary.wiley.com/doi/10.1111/j.1469-185X.2010.00136.x>.
- Matthias Egger, George Davey Smith, Martin Schneider, and Christoph Minder. Bias in meta-analysis detected by a simple, graphical test. *BMJ*, 315:629–634, 9 1997. ISSN 0959-8138. doi: 10.1136/BMJ.315.7109.629. URL <https://www.bmj.com/content/315/7109/629><https://www.bmj.com/content/315/7109/629.abstract>.
- Marco Festa-Bianchet. When does selective hunting select, how can we tell, and what should we do about it? *Mammal Review*, 47:76–81, 1 2017. ISSN 1365-2907. doi: 10.1111/MAM.12078. URL <https://onlinelibrary.wiley.com/doi/10.1111/mam.12078>.
- Joern Fischer and David B. Lindenmayer. Landscape modification and habitat fragmentation: A synthesis. *Global Ecology and Biogeography*, 16:265–280, 5 2007. ISSN 1466822X. doi: 10.1111/J.1466-8238.2007.00287.X. URL <https://onlinelibrary.wiley.com/doi/10.1111/j.1466-8238.2007.00287.x>.
- Jonah Gabry, Rok Češnovar, Andrew Johnson, and Steve Bronder. *cmdstanr: R Interface to 'CmdStan'*, 2025. URL <https://mc-stan.org/cmdstanr/>. R package version 0.9.0, <https://discourse.mc-stan.org>.
- Dany Garant, Loeske E.B. Kruuk, Robin H. McCleery, and Ben C. Sheldon. Evolution in a changing environment: a case study with great tit fledging mass. *The American naturalist*, 164:115–129, 2004. ISSN 15375323. doi: 10.1086/424764.
- P. Gienapp, C. Teplitsky, J. S. Alho, J. A. Mills, and J. Merilä. Climate change and evolution: Disentangling environmental and genetic responses. *Molecular Ecology*, 17:167–178, 1 2008. ISSN 09621083. doi: 10.1111/J.1365-294X.2007.03413.X. URL <https://onlinelibrary.wiley.com/doi/10.1111/j.1365-294X.2007.03413.x>.
- Kiyoko M Gotanda and Lucas D. Gorné. Phenotypic Rates of Change Evolutionary and Ecological Database, 2 2026. URL <https://onlinelibrary.wiley.com/doi/10.1111/mec.16299>.

- Andrew P. Hendry, Thomas J. Farrugia, and Michael T. Kinnison. Human influences on rates of phenotypic change in wild animal populations. *Molecular Ecology*, 17:20–29, 1 2008. ISSN 09621083. doi: 10.1111/J.1365-294X.2007.03428.X;CTYPE:STRING:JOURNAL. URL <https://onlinelibrary.wiley.com/doi/10.1111/j.1365-294X.2007.03428.x>.
- Andrew P. Hendry, Kiyoko M. Gotanda, and Erik I. Svensson. Human influences on evolution, and the ecological and societal consequences. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 372, 1 2017. ISSN 14712970. doi: 10.1098/RSTB.2016.0028/30286.
- Cody E. Hinchliff, Stephen A. Smith, James F. Allman, J. Gordon Burleigh, Ruchi Chaudhary, Lyndon M. Coghill, Keith A. Crandall, Jiabin Deng, Bryan T. Drew, Romina Gazis, Karl Gude, David S. Hibbett, Laura A. Katz, H. Dail Laughinghouse, Emily Jane McTavish, Peter E. Midford, Christopher L. Owen, Richard H. Ree, Jonathan A. Rees, Douglas E. Soltisc, Tiffani Williams, and Karen A. Cranston. Synthesis of phylogeny and taxonomy into a comprehensive tree of life. *Proceedings of the National Academy of Sciences of the United States of America*, 112(41):12764–12769, 10 2015. ISSN 10916490. doi: 10.1073/PNAS.1423041112.
- Daniël Lakens. Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and anovas. *Frontiers in Psychology*, 4:62627, 11 2013. ISSN 16641078. doi: 10.3389/FPSYG.2013.00863. URL www.frontiersin.org.
- Jonathan B. Losos, Kenneth I. Warheit, and Thomas W. Schoener. Adaptive differentiation following experimental island colonization in anolis lizards. *Nature* 1997 387:6628, 387:70–73, 5 1997. ISSN 1476-4687. doi: 10.1038/387070a0. URL <https://www.nature.com/articles/387070a0>.
- Michael Lynch. The rate of morphological evolution in mammals from the standpoint of the neutral expectation. *The American Naturalist*, 136(6):727–741, 1990. ISSN 00030147, 15375323. URL <http://www.jstor.org/stable/2462164>.
- Juha Merilä and Andrew P. Hendry. Climate change, adaptation, and phenotypic plasticity: The problem and the evidence. *Evolutionary Applications*, 7:1–14, 1 2014. ISSN 17524563. doi: 10.1111/EVA.12137. URL <https://onlinelibrary.wiley.com/doi/10.1111/eva.12137>.
- François Michonneau, Joseph W. Brown, and David J. Winter. rotl: an R package to interact with the Open Tree of Life data. *Methods in Ecology and Evolution*, 7(12):1476–1481, 12 2016. ISSN 2041210X. doi: 10.1111/2041-210X.12593;JOURNAL:JOURNAL:2041210X. URL <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.12593>.

Shinichi Nakagawa, Malgorzata Lagisz, Michael D. Jennions, Julia Koricheva, Daniel W.A. Noble,
 Timothy H. Parker, Alfredo Sánchez-Tójar, Yefeng Yang, and Rose E. O’Dea. Methods for testing
 publication bias in ecological and evolutionary meta-analyses. *Methods in Ecology and Evolution*, 13:
 4–21, 1 2022. ISSN 2041210X. doi: 10.1111/2041-210X.13724;REQUESTEDJOURNAL:JOURNAL:
 2041210X;WEBSITE:WEBSITE:BESJOURNALS;WGROU:STRING:PUBLICATION. URL [/doi/
 pdf/10.1111/2041-210X.13724https://onlinelibrary.wiley.com/doi/abs/10.1111/2041-210X.
 13724https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210X.13724](https://onlinelibrary.wiley.com/doi/abs/10.1111/2041-210X.13724).

Shinichi Nakagawa, Ayumi Mizuno, Kyle Morrison, Lorenzo Ricolfi, Coralie Williams, Szymon M. Dro-
 bniak, Malgorzata Lagisz, and Yefeng Yang. Location-Scale Meta-Analysis and Meta-Regression as
 a Tool to Capture Large-Scale Changes in Biological and Methodological Heterogeneity: A Spot-
 light on Heteroscedasticity. *Global Change Biology*, 31(5):e70204, 5 2025a. ISSN 1365-2486. doi:
 10.1111/GCB.70204. URL <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70204>.

Shinichi Nakagawa, Ayumi Mizuno, Coralie Williams, Santiago Ortega, Szymon M. Drobnia, Malgorzata
 Lagisz, Yefeng Yang, Alistair M. Senior, Daniel W. A. Noble, and Erick Lundgren. Mastering an
 Accurate and Generalizable Simulation-Based Method to Obtain Bias-corrected Point Estimates and
 Sampling Variance for Any Effect Sizes. 10 2025b. URL <https://arxiv.org/pdf/2510.27519>.

Shinichi Nakagawa, Ayumi Mizuno, Coralie Williams, Malgorzata Lagisz, Rose E O’Dea, Daniel W.A. Noble,
 Alistair M Senior, Erick Lundgren, and Santiago Ortega. A new effect size for meta-analysis of magni-
 tude: lnM. 2 2026a. doi: 10.32942/X24666. URL <https://ecoevorxiv.org/repository/view/11886/>.

Shinichi Nakagawa, Santiago Ortega, Ayumi Mizuno, Eduardo S.A. Santos, Malgorzata Lagisz, Bhavya Jain,
 Jimuel Jr Celeste, and Sergio Poo Hernandez. *prepR4pcm: Prepare Data and Trees for Phylogenetic
 Comparative Methods*, 2026b. URL <https://github.com/itchyshin/prepR4pcm>. R package version
 1.0.0.

Rose E. O’Dea, Malgorzata Lagisz, Michael D. Jennions, Julia Koricheva, Daniel W.A. Noble, Timothy H.
 Parker, Jessica Gurevitch, Matthew J. Page, Gavin Stewart, David Moher, and Shinichi Nakagawa.
 Preferred reporting items for systematic reviews and meta-analyses in ecology and evolutionary biology:
 a PRISMA extension. *Biological Reviews*, 96(5):1695–1722, 10 2021. ISSN 1469185X. doi: 10.1111/brv.
 12721. URL <https://onlinelibrary.wiley.com/doi/10.1111/brv.12721>.

Santiago Ortega, Shinichi Nakagawa, Eduardo S A Santos, Kiyoko M Gotanda, Sarah Sanderson, Lucas D
 Gorné, and Andrew Hendry. Human disturbance and the magnitude of phenotypic divergence, May
 2026. URL osf.io/9fq4n.

- Arpat Ozgul, Dylan Z. Childs, Madan K. Oli, Kenneth B. Armitage, Daniel T. Blumstein, Lucretia E. Olson, Shripad Tuljapurkar, and Tim Coulson. Coupled dynamics of body mass and population growth in response to environmental change. *Nature* 2010 466:7305, 466:482–485, 7 2010. ISSN 1476-4687. doi: 10.1038/nature09210. URL <https://www.nature.com/articles/nature09210>.
- Eric P. Palkovacs, Michael T. Kinnison, Cristian Correa, Christopher M. Dalton, and Andrew P. Hendry. Fates beyond traits: Ecological consequences of human-induced trait change. *Evolutionary Applications*, 5:183–191, 2 2012. ISSN 17524563. doi: 10.1111/J.1752-4571.2011.00212.X. URL <https://onlinelibrary.wiley.com/doi/10.1111/j.1752-4571.2011.00212.x>.
- Emmanuel Paradis, Julien Claude, and Korbinian Strimmer. Ape: Analyses of phylogenetics and evolution in r language. *Bioinformatics*, 20:289–290, 1 2004. ISSN 1367-4803. doi: 10.1093/BIOINFORMATICS/BTG412.
- Thomas Pendersen. patchwork: The composer of plots, 2025.
- R Core Team. R: A language and environment for statistical computing, 6 2026.
- David N. Reznick, Frank H. Shaw, F. Helen Rodd, and Ruth G. Shaw. Evaluation of the rate of evolution in natural populations of guppies (*poecilia reticulata*). *Science*, 275:1934–1937, 3 1997. ISSN 00368075. doi: 10.1126/SCIENCE.275.5308.1934.
- Simone Des Roches, David M. Post, Nash E. Turley, Joseph K. Bailey, Andrew P. Hendry, Michael T. Kinnison, Jennifer A. Schweitzer, and Eric P. Palkovacs. The ecological importance of intraspecific variation. *Nature Ecology & Evolution* 2017 2:1, 2:57–64, 12 2017. ISSN 2397-334X. doi: 10.1038/s41559-017-0402-5. URL <https://www.nature.com/articles/s41559-017-0402-5>.
- Sarah Sanderson, Marc Olivier Beausoleil, Rose E. O’Dea, Zachary T. Wood, Cristian Correa, Victor Frankel, Lucas D. Gorné, Grant E. Haines, Michael T. Kinnison, Krista B. Oke, Fanie Pelletier, Felipe Pérez-Jvostov, Winer D. Reyes-Corral, Yanny Ritchot, Freedom Sorbara, Kiyoko M. Gotanda, and Andrew P. Hendry. The pace of modern life, revisited. *Molecular Ecology*, 31(4):1028–1043, 2 2022. ISSN 1365294X. doi: 10.1111/MEC.16299.
- Sarah Sanderson, Daniel I. Bolnick, Michael T. Kinnison, Rose E. O’Dea, Lucas D. Gorné, Andrew P. Hendry, and Kiyoko M. Gotanda. Contemporary changes in phenotypic variation, and the potential consequences for eco-evolutionary dynamics. *Ecology Letters*, 26:S127–S139, 9 2023. ISSN 14610248. doi: 10.1111/ELE.14186. URL <https://onlinelibrary.wiley.com/doi/10.1111/ele.14186>.
- K. E. Smokorowski and T. C. Pratt. Effect of a change in physical structure and cover on fish and fish

habitat in freshwater ecosystems: A review and meta-analysis. *Environmental Reviews*, 15:15–41, 2007.
doi: 10.1139/A06-007.

Stan Development Team. Stan reference manual, 2025.

Sharon Y. Strauss, Jennifer A. Lau, and Scott P. Carroll. Evolutionary responses of natives to introduced species: What do introductions tell us about natural communities? *Ecology Letters*, 9:357–374, 3 2006. ISSN 14610248. doi: 10.1111/J.1461-0248.2005.00874.X. URL <https://onlinelibrary.wiley.com/doi/10.1111/j.1461-0248.2005.00874.x>.

Monica G. Turner. Disturbance and landscape dynamics in a changing world. *Ecology*, 91(10):2833–2849, 2010. doi: 10.1890/10-0097.1.

Aki Vehtari, Andrew Gelman, Daniel Simpson, Bob Carpenter, and Paul Christian Burkner. Rank-normalization, folding, and localization: An improved $\hat{r}$ for assessing convergence of mcmc (with discussion). <https://doi.org/10.1214/20-BA1221>, 16:667–718, 6 2021. ISSN 1936-0975. doi: 10.1214/20-BA1221.

Hadley Wickham. ggplot2. *WIREs Computational Statistics*, 3:180–185, 3 2011. ISSN 1939-5108. doi: 10.1002/wics.147. URL <https://wires.onlinelibrary.wiley.com/doi/10.1002/wics.147>.

Ke Yang, Enxuan Lin, Wangli Xu, Liping Zhu, and Tiejun Tong. An alternative measure for quantifying the heterogeneity in meta-analysis. *Statistics in Medicine*, 44:e70089, 5 2025. ISSN 10970258. doi: 10.1002/SIM.70089. URL <https://onlinelibrary.wiley.com/doi/10.1002/sim.70089>.